

Giant Axonal Neuropathy

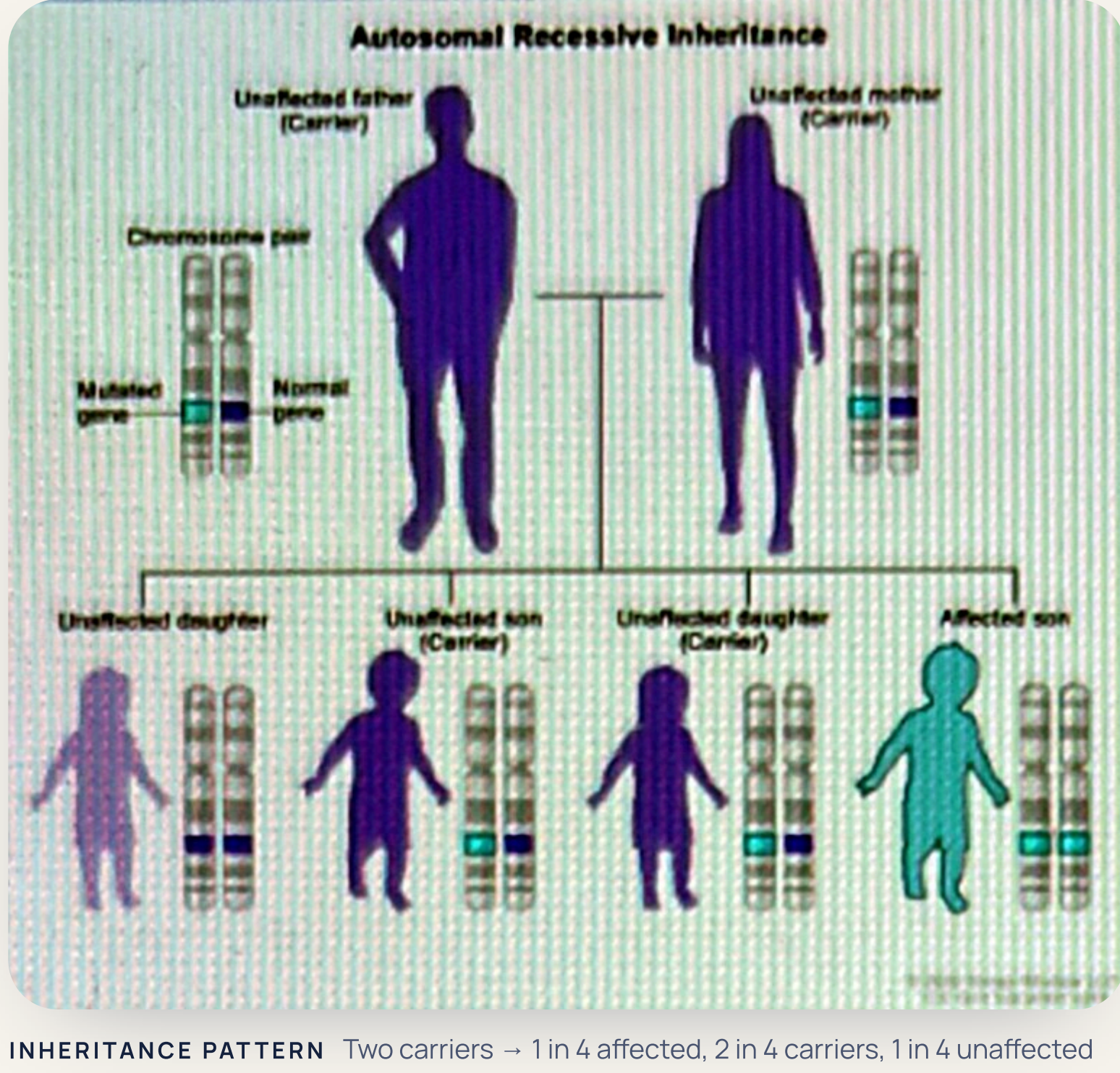
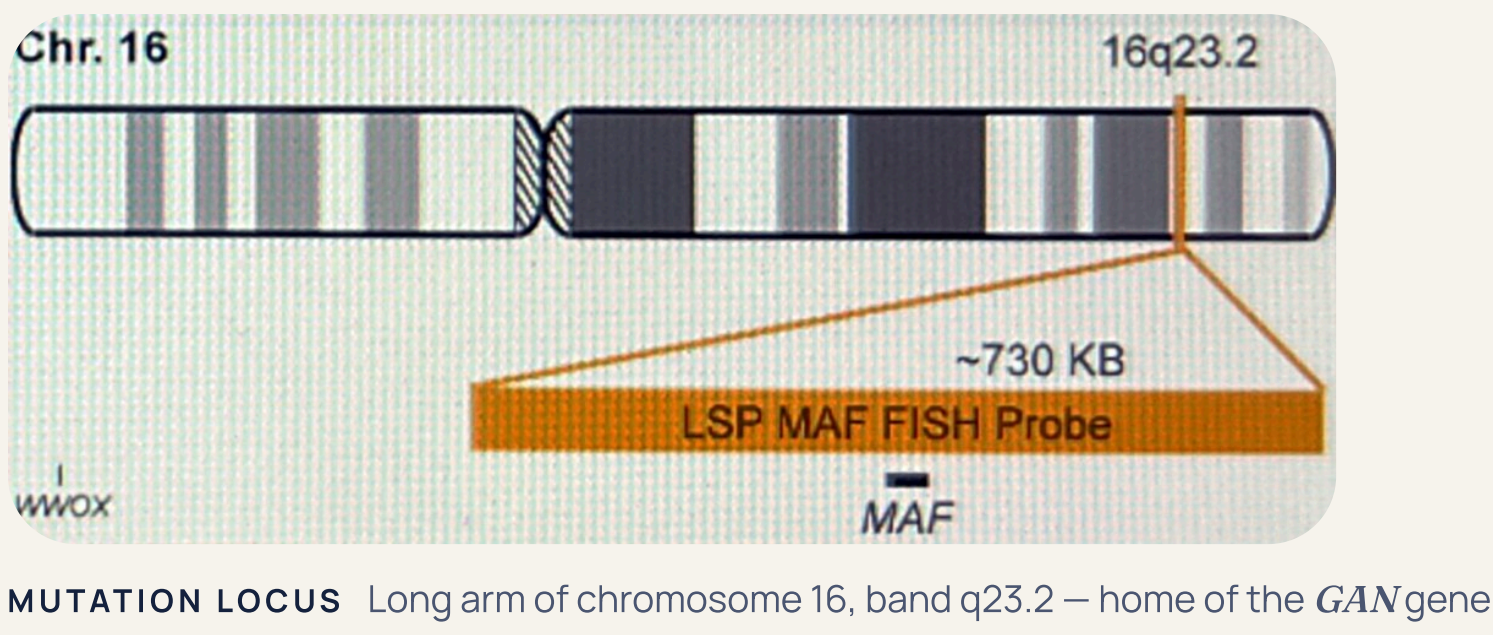
How the loss of one small protein, gigaxonin, makes nerve fibres swell and die — and how one family's fight produced the world's first intrathecal gene therapy.

GENE: **GAN** CHROMOSOME **16Q23.2** AUTOSOMAL RECESSIVE ONSET ≈ **3 YEARS** PREVALENCE < **1 IN 1,000,000**

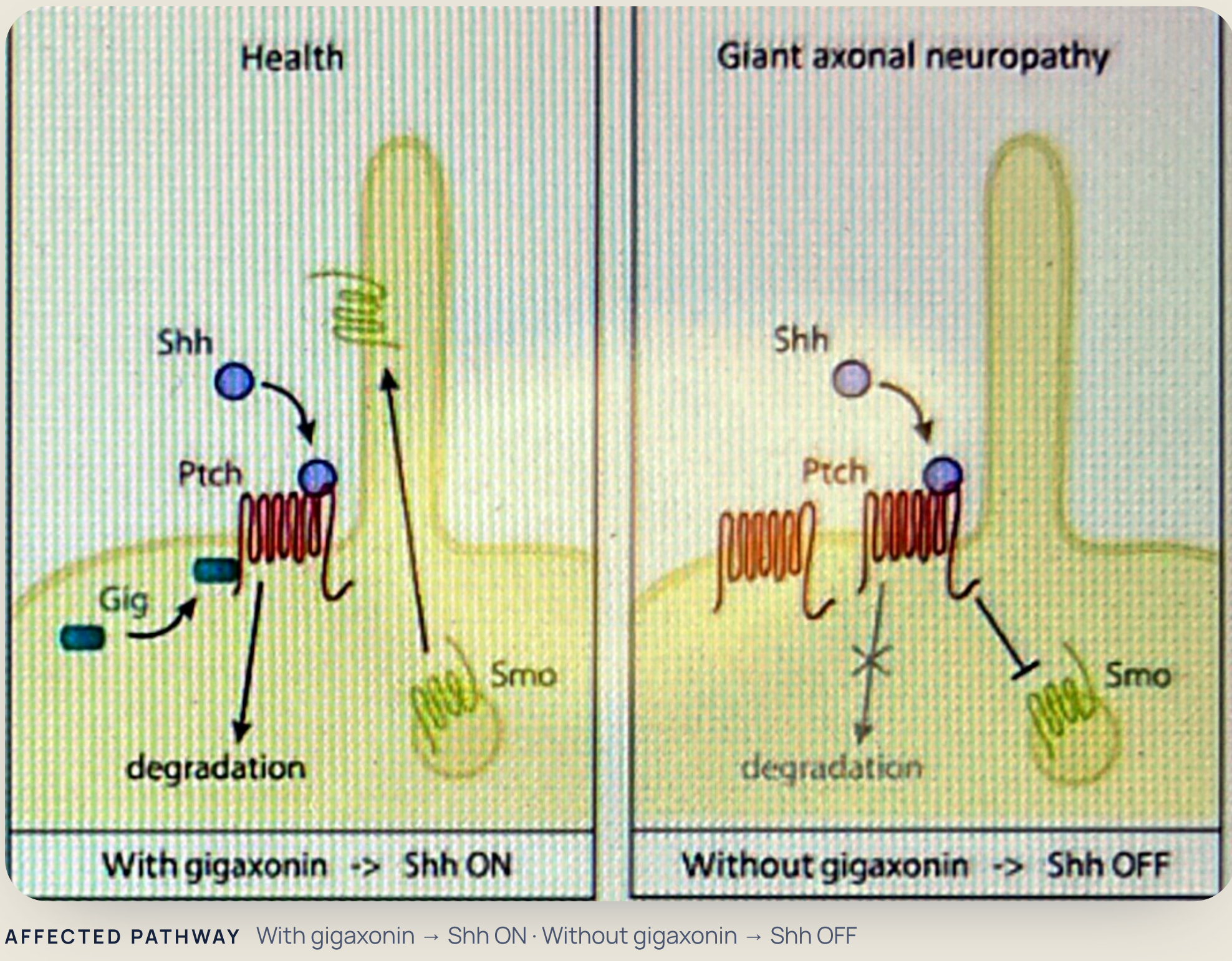
1 One gene, two broken copies

GENE	GAN — chromosome 16, band 16q23.2
PROTEIN	Gigaxonin — 597 aa, BTB/Kelch family
FUNCTION	Adaptor for a Cul3 E3 ubiquitin ligase
MECHANISM	Loss of function — > 100 variants

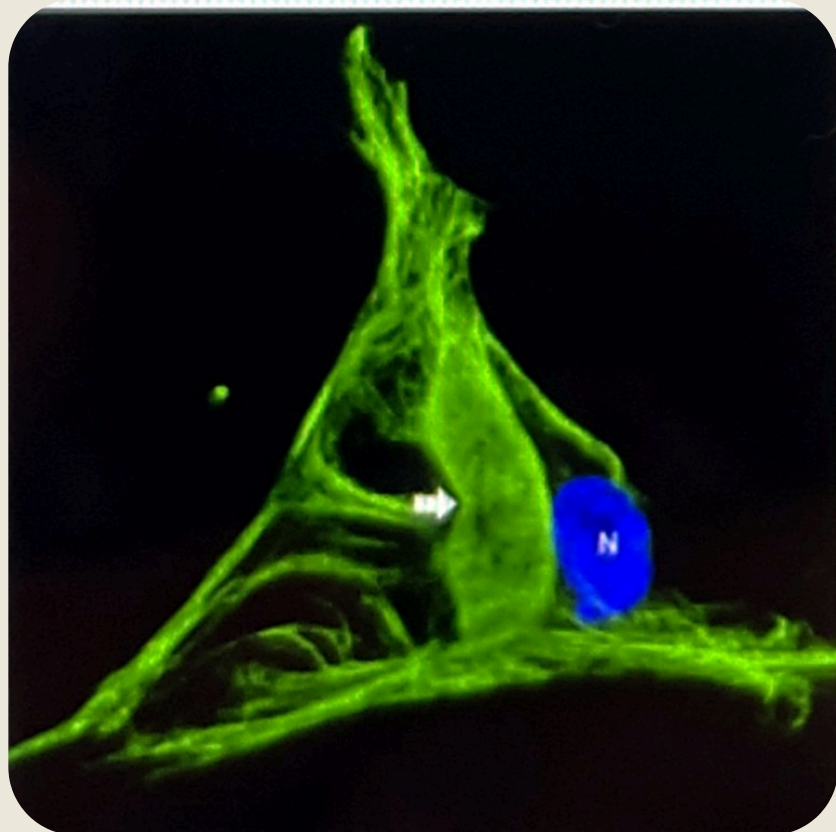
- **Autosomal recessive:** disease appears only when both inherited copies of *GAN* are defective.
- **Carrier parents are healthy** — one working copy is enough. Each pregnancy carries a 25% risk of an affected child.



2 The cell's filament recycler goes missing



- **Normal role:** gigaxonin tags **intermediate filament (IF)** proteins — neurofilaments, vimentin, peripherin, GFAP, keratins — for degradation by the proteasome.
- **Without gigaxonin:** filaments are never cleared and pile up into dense aggregates that choke axonal transport.
- **Second hit:** gigaxonin also degrades the receptor **Patched**. When Patched accumulates it keeps **Smoothed** off, so **Sonic hedgehog signalling stays OFF**.



INTERMEDIATE FILAMENT AGGREGATE
Patient cell — the hallmark of gigaxonin loss (arrow). N = nucleus.

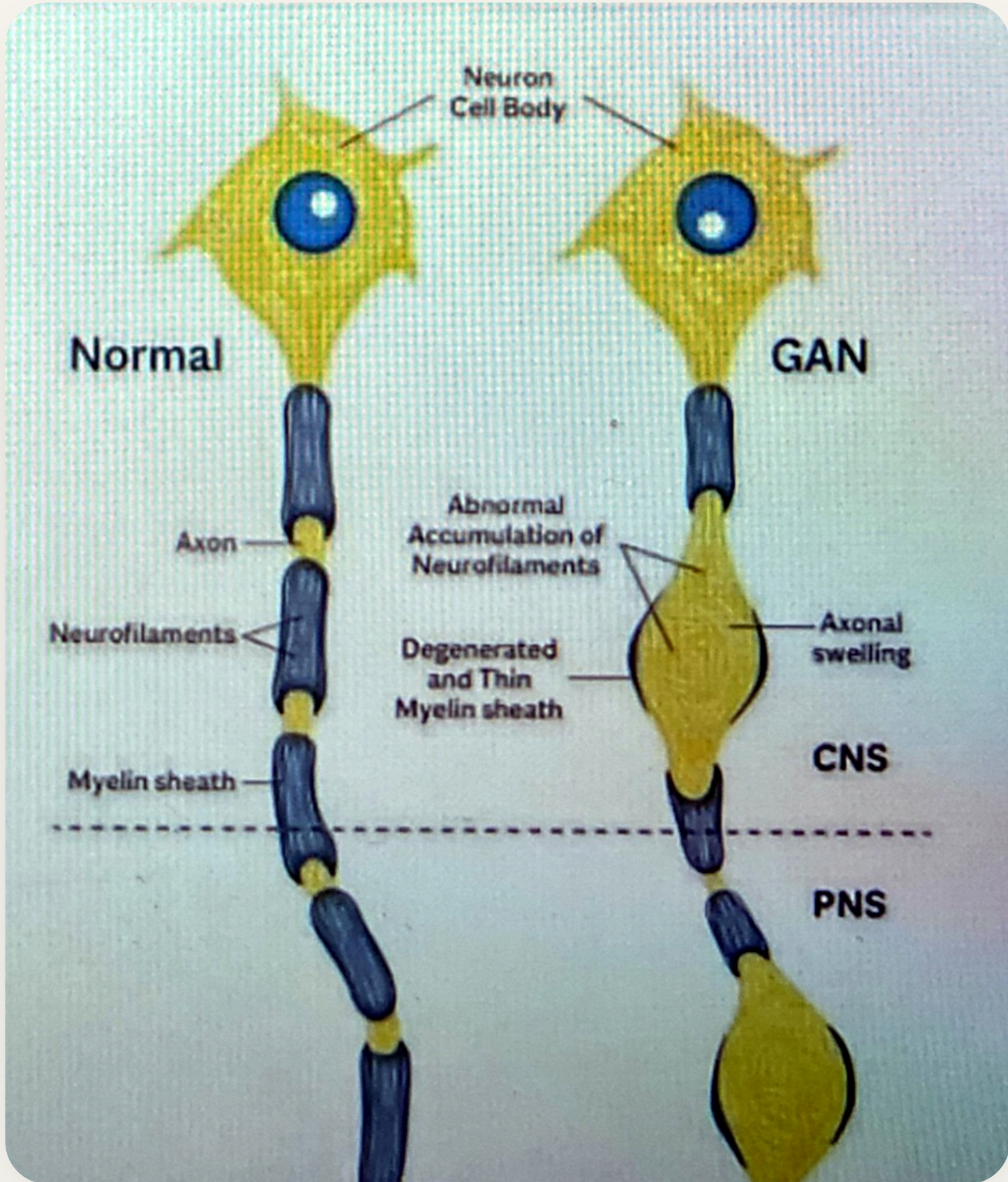
3 Axons that swell into *giants*

- **Focal axonal swellings** several times the normal diameter, packed with tangled neurofilaments — the "giant axons" that name the disease.
- **Myelin thins and degenerates** over each swelling; conduction slows, then fails.
- **Both nervous systems are hit:** peripheral nerves (PNS) and the cerebellum, brainstem and white-matter tracts (CNS).
- **Not only neurons:** GFAP aggregates in astrocytes (Rosenthal fibres); disordered keratin gives the characteristic tightly curled "kinky" hair.

3–5×
typical diameter of a swollen "giant" axon vs. a healthy one

PNS + CNS
a length-dependent axonopathy that becomes a whole-brain disease

MRI
cerebellar and brainstem white-matter change supports diagnosis



NORMAL VS GAN Neurofilament accumulation → axonal swelling → myelin loss, in CNS and PNS

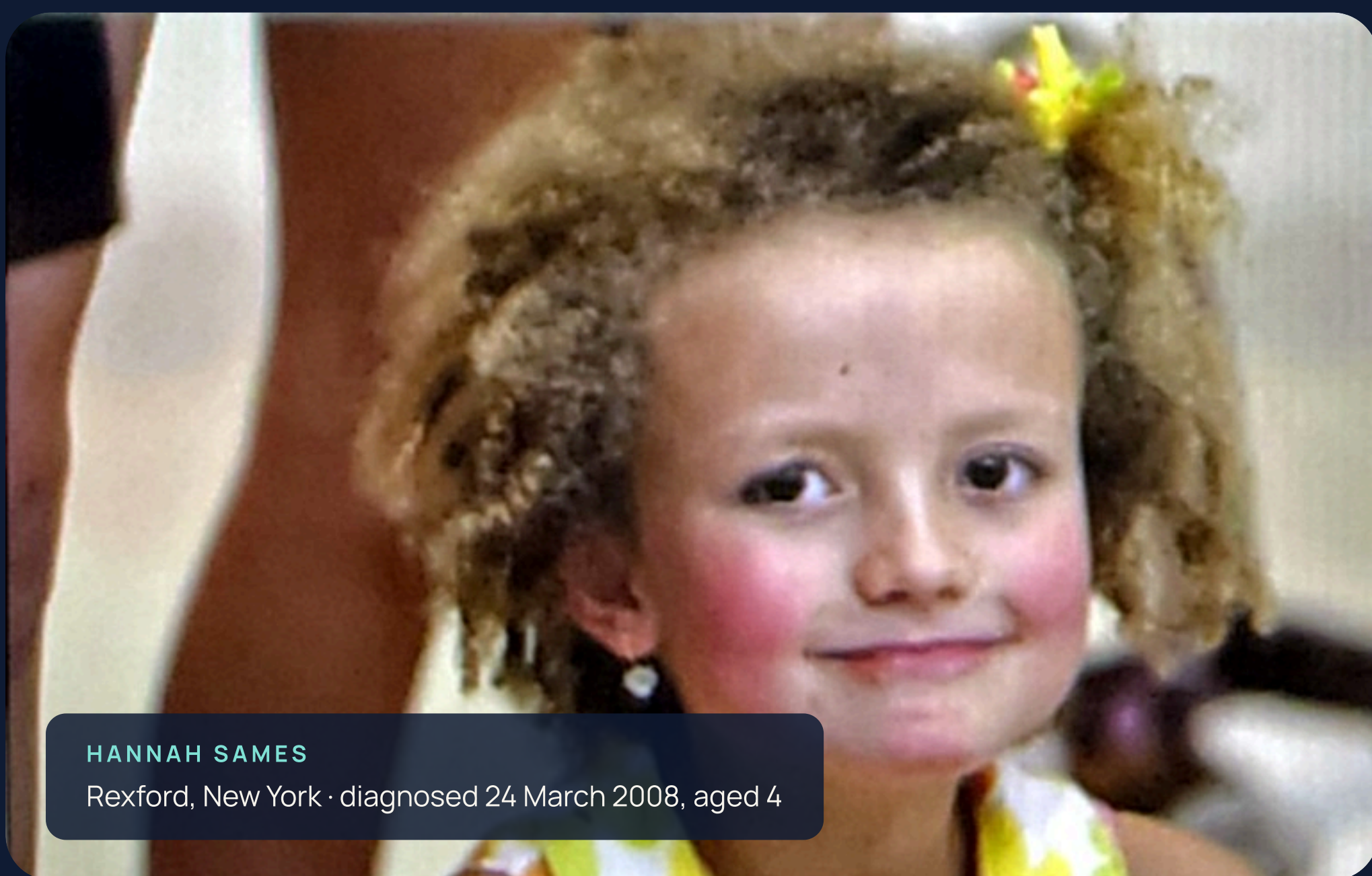
4 A slow, relentless loss — from the feet upward

ONSET - ≈ 3 YEARS
The clumsy walker
Unsteady gait, frequent falls, weak feet and hands, high-arched feet, diminished reflexes. Tightly curled hair unlike either parent is a visible clue.

CHILDHOOD - 5-10 YEARS
Peripheral neuropathy
Loss of vibration and position sense, absent reflexes, muscle wasting below knees and elbows. Nerve biopsy shows giant axons; genetic testing confirms *GAN* variants.

ADOLESCENCE
Central nervous system
Wheelchair dependence, cerebellar ataxia, nystagmus, slurred speech, scoliosis, seizures and cognitive decline as brain white matter degenerates.

EARLY ADULTHOOD
Late stage
Swallowing and breathing muscles weaken; most patients die of respiratory failure in the 2nd–3rd decade. Until 2015 there was **no treatment** beyond supportive care.



HANNAH SAMES
Rexford, New York · diagnosed 24 March 2008, aged 4

THE CASE THAT CHANGED THE FIELD

"We decided to *fight*."

- **2007:** at age 3 Hannah's walking lags behind; a year of tests ends with a GAN diagnosis in **March 2008**.
- **The hardest genotype:** two deletions — Hannah makes **no gigaxonin at all** (CRIM-negative), so her immune system had never "seen" the protein.
- **Hannah's Hope Fund**, founded within months by Lori and Matt Sames; convenes the first GAN symposium (2008), raises **>\$8 million** and funds Steven Gray (UNC) to build the vector.

Hannah's parents found themselves in the unimaginable situation of having funded a clinical trial that might have to exclude their own child.

DNA SCIENCE, JULY 2016

6 THE GENETIC TREATMENT · SCAAV9/JET-GAN · FIRST-IN-HUMAN INTRATHECAL AAV TRIAL, NIH CLINICAL CENTER 2015 (NCT02362438)

A working gene, delivered into the spinal fluid

1 Build the vector
scAAV9: a self-complementary AAV9 shell that enters neurons and spreads through spinal cord and brain. No viral genes; does not integrate.

2 Load the cargo
A **codon-optimised human *GAN*** gene driven by **JeT**, a tiny synthetic promoter that fits the compact genome.

3 Single lumbar injection
One dose — up to **3.5 × 10¹⁴** vector genomes — into the CSF, bathing motor neurons and dorsal root ganglia and bypassing the blood–brain barrier.

4 Restore gigaxonin
Transduced neurons make gigaxonin within ~2 weeks, clearing aggregates and letting damaged sensory nerves **regenerate**.

May 2015
first human ever to receive a therapeutic gene via the spinal fluid — a GAN patient

21 July 2016
Hannah, aged 12, treated as the 5th child (≈120 trillion vector genomes)

CRIM-negative
making no gigaxonin, Hannah needed a dedicated immunosuppression protocol

7 WHAT THE TRIAL SHOWED — NEJM, MARCH 2024

Bending the curve of an "untreatable" disease

14
children aged 6–14 treated at four dose levels, followed for a median of 68.7 months

1
serious adverse event (fever) possibly related; CSF inflammation managed with steroids

–7.2 → –1.9
MFM-32 points lost per year before vs. after the 1.8 × 10¹⁴ vg dose

99%
posterior probability that motor decline slowed at that dose — above the 95% bar

6 / 14
regained, stabilised or newly recordable sensory nerve responses after dosing

3–48×
more regenerating nerve-fibre clusters in sensory-nerve biopsies at one year

- **Not a cure — yet.** Existing damage is not reversed; benefit is largest when children are treated **young and early**.
- **A template for the field:** intrathecal AAV9 delivery pioneered here now underpins trials for many other childhood brain diseases.
- **Next:** programme licensed to industry (Taysha, TSHA-120); Hannah's Hope Fund funds a second injection site.

One family with no medical background turned a diagnosis into a first-in-human therapy — proof of what patient-driven science can do.

TAKE-HOME MESSAGE